代数优化

例：Student(S#, Sname, Sex, Class)

Course(C#, Cname, Teacher)

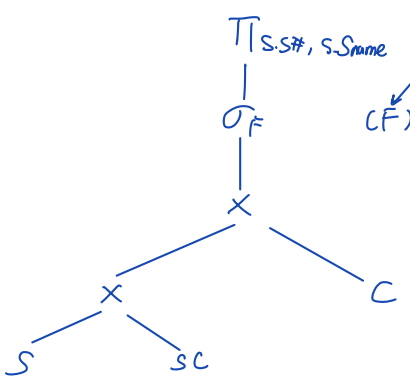
SC(S#, C#, Grade)

查询：找出选修“数据库系统”课程并且成绩为优的男生的学号姓名

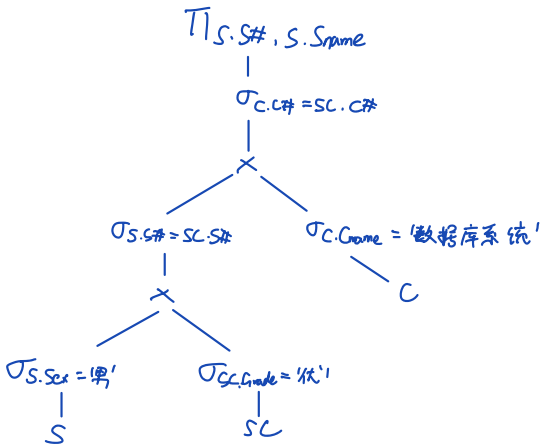
SQL：
```
SELECT S.S#, S.Sname
FROM Student S, Course C, SC
WHERE S.S# = SC.S# AND C.C# = SC.C#
  AND S.Sex = '男' AND C.Cname = '数据库系统'
  AND SC.Grade = '优'
```

关系代数：$\Pi_{S.S\#,\ S.Sname}(\sigma_{S.S\#=SC.S\#\ \wedge\ C.C\#=SC.C\#\ \wedge\ S.Sex='男'\ \wedge\ C.Cname='数据...'\ \wedge\ SC.Grade='优'}(S\times C\times SC))$

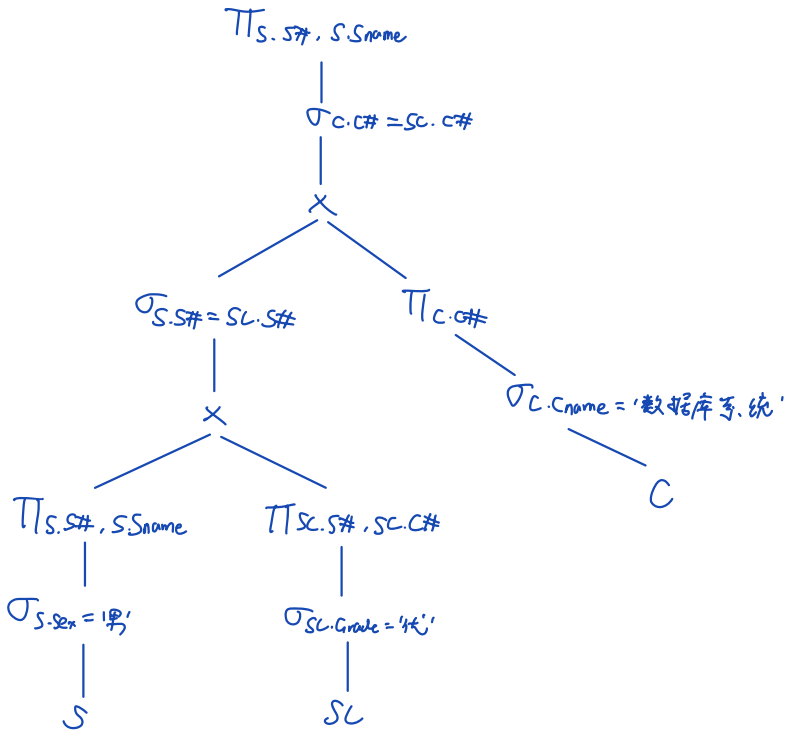
① 原始查询树：



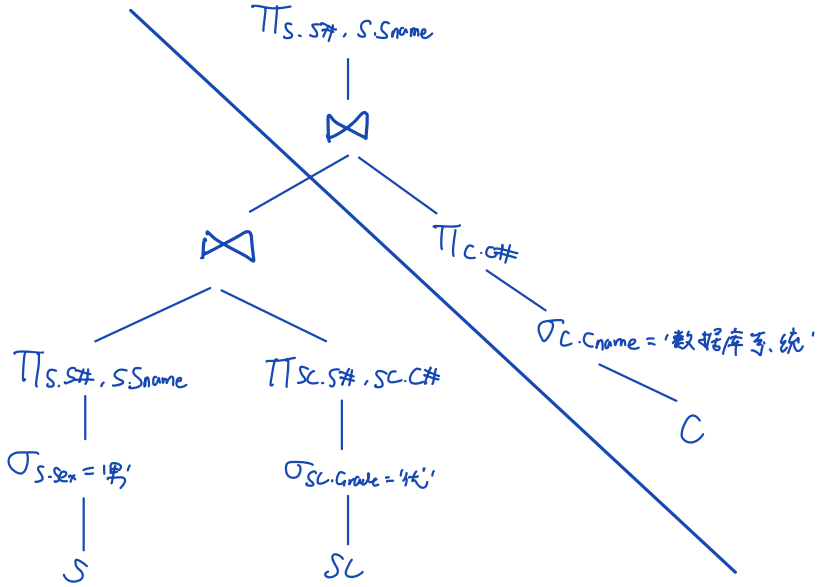
② 选择下推：



③ 投影下推：


$\Pi_{S.S\#,\ S.Sname}$
$\sigma_{C.C\#=SC.C\#}$
×
$\sigma_{S.S\#=SC.S\#}$
$\Pi_{C.C\#}$
×
$\sigma_{C.Cname='数据库系统'}$
C
$\Pi_{S.S\#,\ S.Sname}$
$\Pi_{SC.S\#,\ SC.C\#}$
$\sigma_{S.Sex='男'}$
$\sigma_{SC.Grade<'优'}$
S
SC

④ 转为连接：


$\Pi_{S.S\#,\ S.Sname}$
⋈
⋈
$\Pi_{C.C\#}$
$\sigma_{C.Cname='数据库系统'}$
C
$\Pi_{S.S\#,\ S.Sname}$
$\Pi_{SC.S\#,\ SC.C\#}$
$\sigma_{S.Sex='男'}$
$\sigma_{SC.Grade<'优'}$
S
SC